

# IEOR E4004

## Optimization Models and Methods

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# Outline

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- Solving LPs: Introduction to Simplex Method
  - Matrix Representation of LPs
  - Linear Algebra Basics
  - Changing Forms of LPs
  - The Simplex Method
- Summary

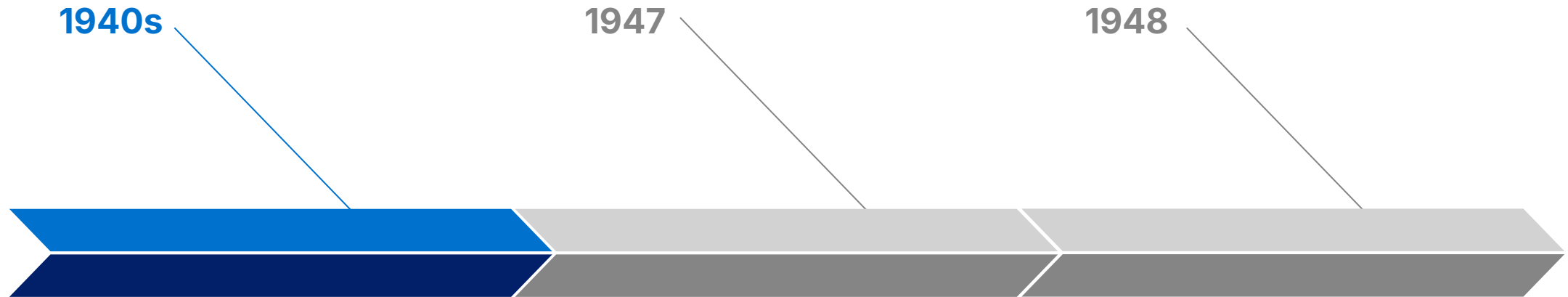
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# **Solving LPs:**

## **Introduction to Simplex Method**

# History of the Simplex Method

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## Development of the Simplex Method

George Dantzig's early work on military logistics leads to the development of the simplex method for solving linear programming problems.

# History of the Simplex Method

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1940s

1947

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## Transformation of the Simplex Algorithm

Dantzig's insight during a lecture by John von Neumann transforms the feasible point method into the systematic simplex algorithm.

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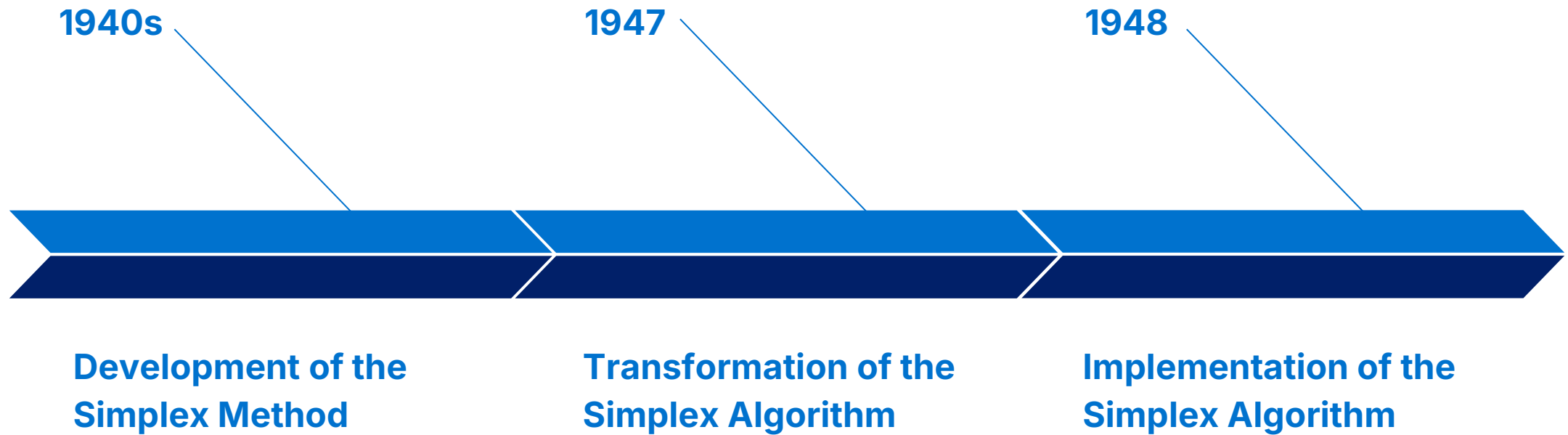
Dantzig's insight during a lecture by John von Neumann transforms the feasible point method into the systematic simplex algorithm.

## Implementation of the Simplex Algorithm

Implementation of the simplex algorithm on the IBM 701 computer demonstrates its practical application and efficiency.

# History of the Simplex Method

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From its wartime origins to modern-day applications,  
the **simplex method** remains a **fundamental optimization tool**.

# The Simplex Method

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- The simplex method is a **powerful algorithm** used to solve linear programming problems

# The Simplex Method

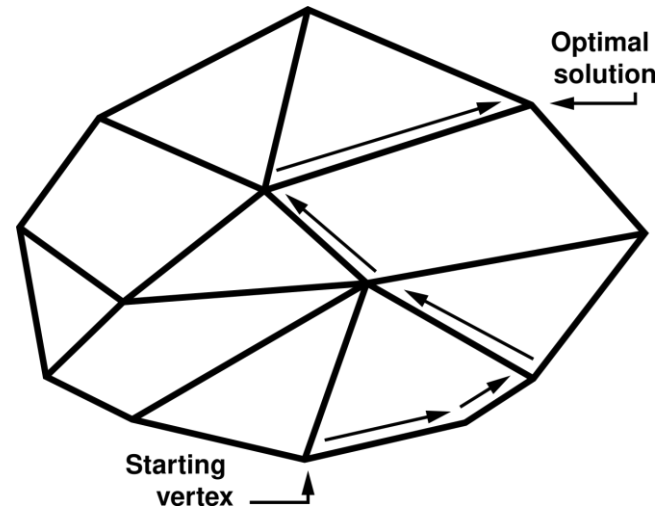
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- The method **systematically moves along the edges of the feasible region**, which is defined by the constraints, to find the optimal solution.



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- The simplex method is a **powerful algorithm** used to solve linear programming problems
- Can be used to solve small LP problems **by hand**
- The method **systematically moves along the edges of the feasible region**, which is defined by the constraints, to find the optimal solution.
- This **iterative process** continues until no further improvements are possible, resulting in the optimal solution.

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# **Matrix Representation and Standard Form of LPs**

# Outline

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- Recap of Basic Linear Algebra Concepts
- Matrix Notation for LPs
- Standard Form of LPs

# Matrices and Linear Systems

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- A **matrix** is any rectangular array of numbers

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- If a matrix **A** has  $m$  rows and  $n$  columns, we call **A** an  $m \times n$  matrix.  
We refer to  $m \times n$  as the **order of the matrix**.

# Matrices and Linear Systems

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- Any matrix with only one column (that is, any  $m \times 1$  matrix) may be thought of as a **column vector**

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- Any vector with only one row (a  $1 \times n$  matrix) as a **row vector**
- Here  $u$ ,  $v$ , and  $w$  are vectors shown as ***directed line segments***

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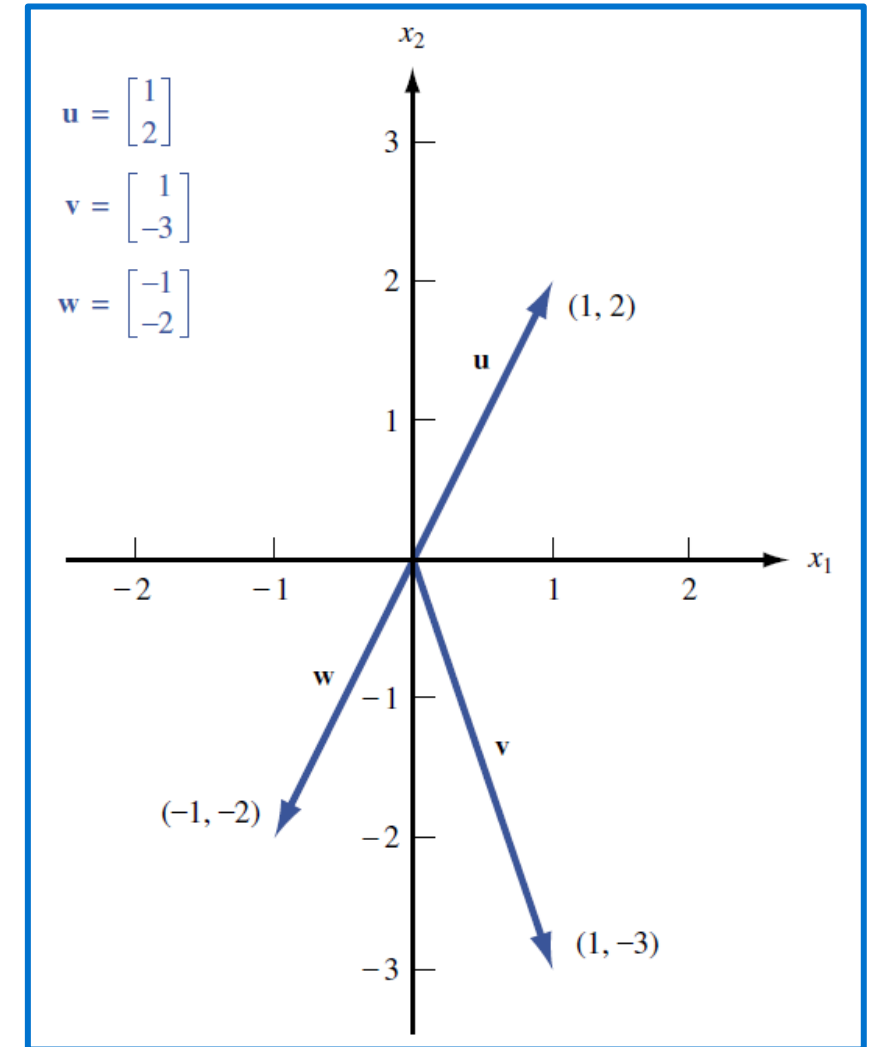
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$$\begin{bmatrix} 0 & 0 \end{bmatrix}$$



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# **Matrices and LPs | Matrix Operations**

# Matrix Operations

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- The Scalar Multiple of a Matrix

$$\text{if } A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \quad \text{then } 3A = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$$

# Matrix Operations

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## ● The Scalar Multiple of a Matrix

$$\text{if } A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \quad \text{then } 3A = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$$

## ● Addition of Two Matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & -2 & -3 \\ 2 & 1 & -1 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1 - 1 & 2 - 2 & 3 - 3 \\ 0 + 2 & -1 + 1 & 1 - 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

# Matrix Operations

---

- The Transpose of a Matrix

$$\text{if } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \text{then } A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

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## ● Matrix Multiplication

$$\text{Compute } C = AB \text{ for } A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}$$

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Because **A** is a  $2 \times 3$  matrix and **B** is a  $3 \times 2$  matrix, **AB** is defined, and **C** will be a  $2 \times 2$  matrix.

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$$c_{11} = [1 \quad 1 \quad 2] \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 1(1) + 1(2) + 2(1) = 5$$

$$c_{12} = [1 \quad 1 \quad 2] \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = 1(1) + 1(3) + 2(2) = 8$$

$$c_{21} = [2 \quad 1 \quad 3] \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 2(1) + 1(2) + 3(1) = 7$$

$$c_{22} = [2 \quad 1 \quad 3] \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = 2(1) + 1(3) + 3(2) = 11$$

$$C = AB = \begin{bmatrix} 5 & 8 \\ 7 & 11 \end{bmatrix}$$

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The matrix product of **A** and **B** (written **AB**) is defined if and only if: Number of columns in **A** = Number of rows in **B**

$x$  2 matrix.

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# Matrices and LPs | Linear Dependence

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For example,  $V = \{[0 \ 0], [1 \ 0], [0 \ 1]\}$  is linearly dependent because if, say,  $c_1 = 0$ , then:

$$c_1([0 \ 0]) + 0([1 \ 0]) + 0([0 \ 1]) = [0 \ 0]$$

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*There is a nontrivial linear combination of vectors in  $V$  that adds up to 0.*

# Linear Independence

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For example,  $V = \{[1 \ 0], [0 \ 1]\}$  is linearly independent because the only linear combination of vectors in  $V$  that yields  $\mathbf{0}$  is the trivial linear combination,  $[c_1 \ c_2] = [0 \ 0]$ , where:

$$c_1([1 \ 0]) + c_2([0 \ 1]) = [0 \ 0]$$

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# **Matrices and LPs | Rank of a Matrix**

# The Rank of a Matrix

---

- Let  $\mathbf{A}$  be any  $m \times n$  matrix, and denote the rows of  $\mathbf{A}$  by  $r_1, r_2, \dots, r_m$  and define  $\mathbf{R} = \{r_1, r_2, \dots, r_m\}$  .

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? What are the ranks of these matrices

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---

# **Matrices and LPs | Linear Algebra with Matrix**

# The Inverse of a Matrix

---

- For a given  $m \times m$  matrix  $\mathbf{A}$ , the  $m \times m$  matrix  $\mathbf{B}$  is the inverse of  $\mathbf{A}$  (denoted as  $\mathbf{A}^{-1}$ ) if  $\mathbf{BA} = \mathbf{AB} = \mathbf{I}_m$ , where  $\mathbf{I}_m$  is the identity matrix of order  $m \times m$ .

# The Determinant of a Matrix

---

The determinant is a scalar value that can be computed from the elements of a square matrix and encodes certain properties of the linear transformation represented by the matrix – will be helpful in non-linear programming.

For a  $1 \times 1$  matrix  $A = [a_{11}]$ ,

$$\det A = a_{11}$$

For a  $2 \times 2$  matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det A = a_{11}a_{22} - a_{21}a_{12}$$

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For  
example:

$$\det \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} = 2(5) - 3(4) = -2$$

# Matrix Operations Using Python

---

Using the “numpy” library, we can find the determinants and inverse of matrices

- Install and import numpy:

```
pip install numpy  
import numpy as np
```

- Declare a matrix:

```
matrix = np.array([[1, 2], [3, 4]])
```

- Find the inverse:

```
inverse_matrix = np.linalg.inv(matrix)
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- Find the determinant:

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**Note:** *Ensure matrices have an inverse before attempting to find one (determinant should not be zero).  
Always check if the matrix is square before finding the determinant or inverse.*

# Linear Systems

---

Consider this system of linear equations:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

Here,  $x_1, x_2, \dots, x_n$  are **variables**, and the  $a_{ij}$  's and  $b_i$  's are **constants**.

# Linear Systems

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These set of equations is called a linear system of  $m$  equations in  $n$  variables.

# Matrix Notation for LPs

---

$$\begin{aligned} & \max/\min \sum_{j=1}^n c_j x_j \\ \text{s.t. } & \sum_{j=1}^n a_{ij} x_j \begin{pmatrix} \leq \\ \geq \\ = \end{pmatrix} b_i, \quad i = 1, \dots, m \\ & x_j \begin{pmatrix} \geq 0 \\ \leq 0 \\ \text{u.r.s.} \end{pmatrix}, \quad j = 1, \dots, n. \end{aligned}$$

There are:

- **$n$**  decision variables,  $x_1, \dots, x_n$
- **$m$**  functional constraints
- The constraints can use a mixture of " $\leq$ ", " $\geq$ ", and " $=$ ".
- Each variable may have the bound  $x_j \geq 0$ ,  $x_j \leq 0$ , or no bound, which we call unrestricted in sign (urs)

# Matrix Notation for LPs

- ! Using matrices can **greatly simplify** the statement and solution of a system of linear equations.

- The linear system

$A\mathbf{x} = \mathbf{b}$ , where:

$$\begin{array}{cccc} a_{11}x_1 & + & a_{12}x_2 & + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + \cdots + a_{2n}x_n = b_2 \\ \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + \cdots + a_{mn}x_n = b_m \end{array}$$

can be represented as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

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 *a.k.a. matrix representation*

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

# The Inverse of a Matrix & Linear Systems

- For a given  $m \times m$  matrix  $A$ , the  $m \times m$  matrix  $B$  is the inverse of  $A$  (denoted as  $A^{-1}$ ) if  $BA = AB = I_m$ , where  $I_m$  is the identity matrix of order  $m \times m$ .

! The concept of a matrix inverse can be used to solve a linear system  $Ax = b$  that has  $m$  equations and  $m$  unknown. Suppose that  $A^{-1}$  exists.

$$(A^{-1}A)\mathbf{x} = A^{-1}\mathbf{b}$$

or  $I_m\mathbf{x} = A^{-1}\mathbf{b}$

or  $\mathbf{x} = A^{-1}\mathbf{b}$

# The Inverse of a Matrix

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$$(\mathbf{A}^{-1}\mathbf{A})\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

or 
$$\mathbf{I}_m\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

or 
$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

If  $m \times m$  matrix  $\mathbf{A}$  has rank  $\mathbf{A} < m$ , then  $\mathbf{A}^{-1}$  will not exist!

# Example | Representing LPs Using Matrix Notation

Consider the linear program

$$\begin{aligned} \min \quad & 9x_1 + 12x_2 + 6x_3 \\ \text{s.t.} \quad & 3x_1 + x_2 - 2x_3 \geq 4 \\ & x_1 + 2x_2 + 4x_3 \geq 6 \\ & x_j \geq 0. \end{aligned}$$

Converting the objective function and constraints to matrix form, we have:

$$\begin{aligned} \min \quad & [9 \quad 12 \quad 6] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} 3 & 1 & -2 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 4 \\ 6 \end{bmatrix} \\ & x_j \geq 0. \end{aligned}$$

# Example | Representing LPs Using Matrix Notation

---

If we let

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 2 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 9 \\ 12 \\ 6 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

then this linear program can be written as:

$$\begin{aligned} \min \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \geq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

**Step 2:** Convert to Standard Linear Programming Form

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

**Step 2:** Convert to Standard Linear Programming Form

**Step 3:** Initialize the Simplex Tableau

# The Simplex Method in a Nutshell

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**Step 4:** Start at a corner of the feasible region

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

**Step 2:** Convert to Standard Linear Programming Form

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**Step 5:** Follow the path and look for an edge that takes you closer to your goal

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

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**Step 4:** Start at a corner of the feasible region

**Step 5:** Follow the path and look for an edge that takes you closer to your goal

**Step 6:** Stop when you can't find an edge that improves your goal! ***You're now at the best corner - the optimal solution!***

# The Simplex Method in a Nutshell

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**Step 1:** Formulate the Linear Programming Problem

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**Step 3:** Initialize the Simplex Tableau

**Step 4:** Start at a corner of the feasible region

**Step 5:** Follow the path and look for an edge that takes you closer to your goal

**Step 6:** Stop when you can't find an edge that improves your goal! ***You're now at the best corner - the optimal solution!***



Understand the solution:

Interpret what your optimal solution means for your real-world situation!

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# The Simplex Method

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# **Standard Form of LPs**

# Standard Form of an LP & Changing Forms

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An LP is in standard form if the following are true:

- All variables have non-negativity constraints,
- All other constraints are strict equality constraints,
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**Is below LP in standard form?**

$$\begin{array}{ll}\max & z = 3x_1 + 2x_2 - x_3 + x_4 \\ \text{s. t.} & x_1 + 2x_2 + x_3 - x_4 \leq 5 \\ & -2x_1 - 4x_2 + x_3 + x_4 \leq -1 \\ & x_1, x_2 \geq 0\end{array}$$

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**✗ No.**

# Changing Forms of LPs

---

**! Any LP can be transformed into an LP in standard form!**

We will learn how to deal with:

- Non-positive decision variables,
- Unrestricted variables,
- Inequalities, and
- Minimization objective

# Dealing with Non-Positive Variables ( $x \leq 0$ )

---

Let's look at the LP below:

$$\begin{aligned}\max Z &= -x_1 - 2x_2 - x_3 + x_4 \\ x_1 + 3x_2 + 4x_3 - x_4 &\leq 5 \\ x_1 \geq 0, x_2 \leq 0, x_3 \leq 0, x_4 &\geq 0\end{aligned}$$

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Define  $x_2$  and  $x_3$  as new non-negative variables, where  $x_2 = -y_2$  and  $x_3 = -y_3$

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Then, the LP becomes

$$\begin{aligned}\max Z &= -x_1 + 2y_2 + y_3 + x_4 \\ x_1 - 3y_2 - 4y_3 - x_4 &\leq 5 \\ x_1 &\geq 0, y_2 \geq 0, y_3 \geq 0, x_4 \geq 0\end{aligned}$$

# Dealing with Variables Unrestricted in Sign

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Consider the following LP:

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The new LP:

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Replace it by a pair of new non-negative variables,  $x_2^+$  and  $x_2^-$ , where  $x_2 = x_2^+ - x_2^-$

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The new LP:

$$\begin{aligned}\max \quad & Z = -x_1 - 2x_2^+ + 2x_2^- - x_3 + x_4 \\ & x_1 + 3x_2^+ - 3x_2^- + 4x_3 - x_4 \leq 5 \\ & x_1 \geq 0, x_2^+ \geq 0, x_2^- \geq 0, x_3 \geq 0, x_4 \geq 0\end{aligned}$$

***Don't forget to convert the original variable by inspection after solution of the new LP!***

# Converting Inequalities to Equalities ( $\leq$ to $=$ )

---

Add a non-negative **slack variable** to the left-hand-side (LHS)

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For example, the following inequality:

$$x_1 + 2x_2 + x_3 - x_4 \leq 5$$

becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 + \mathbf{s_1} = 5 \quad (\text{note we also have } s_1 \geq 0)$$

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becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 + s_1 = 5 \quad (\text{note we also have } s_1 \geq 0)$$

The slack variable measures the amount of unused resource.

So, we can also say that

$$s_1 = 5 - x_1 - 2x_2 - x_3 + x_4$$

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Subtract a non-negative **surplus variable** from the left-hand-side

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becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 - e_1 = 5$$

(note we also have  $e_1 \geq 0$ )

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Subtract a non-negative **surplus variable** from the left-hand-side

For example, the following inequality:

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becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 - e_1 = 5 \quad (\text{note we also have } e_1 \geq 0)$$

The surplus variable measures the amount by which ***we over-satisfy a requirement***.

So, we can also say that:

$$e_1 = 5 - x_1 + 2x_2 + x_3 - x_4$$

# Converting Minimization to Maximization

---

Recognize that to *minimize* " $z$ ", you could alternatively  
*maximize* " $-z$ "

# Converting Minimization to Maximization

Recognize that to *minimize* " $z$ ", you could alternatively *maximize* " $-z$ "

For example, the following minimization objective function:

$$\min x_1 + 2x_2 + x_3 - x_4$$

Can also be written as a maximization problem

$$\max -x_1 - 2x_2 - x_3 + x_4$$

# In-Class Exercise/Quiz Question – Standard Form Conversion

---

Transform the following LP to standard form:

**Minimize**  $Z = x_1 - x_2 + 2x_3 - x_4$

**Subject to:**

$$\begin{array}{rcll} 2x_1 - 3x_2 + x_3 & & \geq & 8 \\ 2x_1 + 2x_2 + 3x_3 - x_4 & \leq & 2 & \\ 4x_1 + 4x_2 + 2x_3 - x_4 & \geq & 6 & \\ x_1 - 2x_2 + 3x_3 - x_4 & \leq & 2 & \\ & x_1 & \geq & 0 \\ & x_2 & \leq & 0 \\ & x_4 & \geq & 0 \end{array}$$

# Solution – Standard Form Conversion

Transform the following LP to standard form:

$$\text{Minimize } Z = x_1 - x_2 + 2x_3 - x_4$$

$$\begin{aligned} \text{Subject to: } & 2x_1 - 3x_2 + x_3 \geq 8 \\ & 2x_1 + 2x_2 + 3x_3 - x_4 \leq 2 \\ & 4x_1 + 4x_2 + 2x_3 - x_4 \geq 6 \\ & x_1 - 2x_2 + 3x_3 - x_4 \leq 2 \\ & x_1 \geq 0 \\ & x_2 \leq 0 \\ & x_4 \geq 0 \end{aligned}$$

$$\text{Maximize } -Z = -x_1 - y_2 - 2x_3^+ + 2x_3^- + x_4$$

$$\begin{aligned} \text{Subject to: } & 2x_1 + 3y_2 + x_3^+ - x_3^- - e_1 = 8 \\ & 2x_1 - 2y_2 + 3x_3^+ - 3x_3^- - x_4 + s_1 = 2 \\ & 4x_1 - 4y_2 + 2x_3^+ - 2x_3^- - x_4 - e_2 = 6 \\ & x_1 + 2y_2 + 3x_3^+ - 3x_3^- - x_4 + s_2 = 2 \\ & x_1, x_3^+, x_3^-, x_4, y_2, e_1, e_2, s_1, s_2 \geq 0 \end{aligned}$$

# An Introductory Example: Simplex Method

---

## Step 1: Formulate the LP

- We will work on this small LP :  
Maximize  $x_1 + x_2$   
subject to  $-x_1 + x_2 \leq 1$   
 $x_1 \leq 3$   
 $x_2 \leq 2$   
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## Step 2: Introduce slack variable $x_3, x_4, x_5$ and represent the model in standard LP form

$$\begin{array}{llllll} \text{maximize} & x_1 & + & x_2 & & \\ \text{subject to} & -x_1 & + & x_2 & + & x_3 & = & 1 \\ & x_1 & & & & + & x_4 & = & 3 \\ & & & x_2 & & & + & x_5 & = & 2 \\ & x_1, x_2, \dots, x_5 & \geq & 0, & & & & & & \end{array}$$

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$$A = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

# An Introductory Example cont'd

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**Step 3:** Express each linear program in the form of a *simplex tableau*

$$\begin{array}{rclcl} x_3 & = & 1 & + & x_1 & - & x_2 \\ x_4 & = & 3 & - & x_1 & & \\ x_5 & = & 2 & & & - & x_2 \\ \hline z & = & & & x_1 & + & x_2 \end{array}$$

# An Introductory Example cont'd

**Step 3:** Express each linear program in the form of a *simplex tableau*

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# An Introductory Example cont'd

**Step 3:** Express each linear program in the form of a *simplex tableau*

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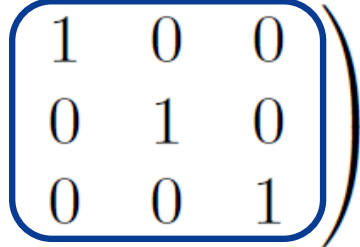

$$\begin{array}{rcll} x_3 & = & 1 & + x_1 - x_2 \\ x_4 & = & 3 & - x_1 \\ x_5 & = & 2 & \phantom{- x_1} - x_2 \\ \hline z & = & & x_1 + x_2 \end{array}$$

# An Introductory Example cont'd

**Step 4:** Select a corner of the feasible region

$$\begin{array}{rcl} x_3 & = & 1 + x_1 - x_2 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 2 \qquad \qquad - x_2 \\ \hline z & = & \qquad x_1 + x_2 \end{array}$$

Identity matrix

$$A = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$


We substitute  $x_1$ , and  $x_2$  with 0 then,  
 $x_3 = 1$ ,  $x_4 = 3$ ,  $x_5 = 2$  and  $z = 0$

# An Introductory Example cont'd

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**Step 4 cont'd:** Select a corner of the feasible region

So,  $x_1 = 0$ ,  $x_2 = 0$ ,  $x_3 = 1$ ,  $x_4 = 3$ ,  $x_5 = 2$  is a Basic Feasible Solution (BFS) where,

$$A = \begin{pmatrix} & x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

# An Introductory Example cont'd

**Step 4 cont'd:** Select a corner of the feasible region

So,  $x_1 = 0$ ,  $x_2 = 0$ ,  $x_3 = 1$ ,  $x_4 = 3$ ,  $x_5 = 2$  is a Basic Feasible Solution (BFS) where,

Non-basic variables (NBV) 

$$A = \begin{pmatrix} -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Basic variables (BV) 

Our Basis is  $\mathbf{B} = \{3, 4, 5\}$

# An Introductory Example cont'd

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**Step 5:** Find the entering variable that will improve  $Z$

Pick a variable arbitrarily from NBVs ( $x_1, x_2$ ) to improve from 0 because both variables have positive coefficients in the objective

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*Let's pick  $x_2$*

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$$\begin{array}{ll} \text{maximize} & x_1 + x_2 \\ \text{subject to} & -x_1 + x_2 + x_3 = 1 \\ & x_1 + x_4 = 3 \\ & x_2 + x_5 = 2 \\ & x_1, x_2, \dots, x_5 \geq 0, \end{array}$$

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How much can you increase  $x_2$ ?

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How much can you increase  $x_2$ ?

*It depends on the feasibility!*

# An Introductory Example cont'd

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**Step 5 cont'd:** Find the entering variable that will improve  $Z$



While increasing  $x_2$ , be careful not to let any of the basic variables  $x_3, x_4, x_5$  go below zero!

# An Introductory Example cont'd

**Step 5 cont'd:** Find the entering variable that will improve  $Z$



While increasing  $x_2$ , be careful not to let any of the basic variables  $x_3, x_4, x_5$  go below zero!

Let's check our BVs:

$$x_3 = 1 + x_1 - x_2$$

$$x_4 = 3 - x_1$$

$$x_5 = 2 - x_2$$

# An Introductory Example cont'd

**Step 5 cont'd:** Find the entering variable that will improve  $z$



While increasing  $x_2$ , be careful not to let any of the basic variables  $x_3, x_4, x_5$  go below zero!

Let's check our BVs:

$$\begin{array}{lcl} x_3 = 1 + x_1 - x_2 & \xrightarrow{\quad} & x_2 = 1 \\ x_4 = 3 - x_1 & & \\ x_5 = 2 - x_2 & \xrightarrow{\quad} & x_2 = 2 \end{array}$$

# An Introductory Example cont'd

**Step 5 cont'd:** Find the entering variable that will improve  $z$



While increasing  $x_2$ , be careful not to let any of the basic variables  $x_3, x_4, x_5$  go below zero!

Let's check our BVs:

$$\begin{array}{rcl} x_3 & = & 1 + x_1 - x_2 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 2 - x_2 \end{array}$$

$x_2 = 1$

$x_2 = 2$

We can only increase  $x_2$  up to 1 without making other variables go below zero!

# An Introductory Example cont'd

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**Step 5 cont'd:** Find the entering variable that will improve  $Z$

The entering variable  $x_2 = 1$  while,  $x_1 = 0$ ,  $x_3 = 0$ ,  $x_4 = 3$ ,  $x_5 = 1$

$$\begin{aligned}x_3 &= 1 + x_1 - x_2 \\x_4 &= 3 - x_1 \\x_5 &= 2 - x_2\end{aligned}$$

# An Introductory Example cont'd

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**Step 5 cont'd:** Find the entering variable that will improve  $z$

The entering variable  $x_2 = 1$  while,  $x_1 = 0$ ,  $x_3 = 0$ ,  $x_4 = 3$ ,  $x_5 = 1$

$$\begin{aligned}x_3 &= 1 + x_1 - x_2 \\x_4 &= 3 - x_1 \\x_5 &= 2 - x_2\end{aligned}$$

***Basis becomes  $B = \{2, 4, 5\}$ , and  $z = 1$***

# An Introductory Example cont'd

**Step 6:** Update the simplex tableau

$$\begin{array}{rclcl} x_2 & = & 1 & + & x_1 & - & x_3 \\ x_4 & = & 3 & - & x_1 & & \\ x_5 & = & 1 & - & x_1 & + & x_3 \\ \hline z & = & 1 & + & 2x_1 & - & x_3 \end{array}$$

The process of rewriting one simplex tableau into another is called a **pivot step**!

# An Introductory Example cont'd

---

**Repeat Step 5:** Find the entering variable that will improve  $z$

The objective becomes  $z = 1 + 2x_1 - x_3$

Only improving  $x_1$  will improve  $z$ ; thus, the entering variable is  $x_1$

# An Introductory Example cont'd

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How much can you increase  $x_1$ ?

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The objective becomes  $z = 1 + 2x_1 - x_3$

Only improving  $x_1$  will improve  $z$ ; thus, the entering variable is  $x_1$

?

How much can you increase  $x_1$ ?

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

Does not restrict!

$x_1 = 3$

$x_1 = 1$

# An Introductory Example cont'd

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**Step 5 cont'd:** Find the entering variable that will improve  $z$

The entering variable  $x_1 = 1$  while,  $x_2 = 2$ ,  $x_3 = 0$ ,  $x_4 = 2$ ,  $x_5 = 0$

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

# An Introductory Example cont'd

**Step 5 cont'd:** Find the entering variable that will improve  $z$

The entering variable  $x_1 = 1$  while,  $x_2 = 2$ ,  $x_3 = 0$ ,  $x_4 = 2$ ,  $x_5 = 0$

*Basis becomes  $B = \{1, 2, 4\}$ , and  $z = 3$*

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

# An Introductory Example cont'd

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**Step 6:** Update the simplex tableau

$$\begin{array}{rclcl} x_1 & = & 1 & + & x_3 & - & x_5 \\ x_2 & = & 2 & & & - & x_5 \\ x_4 & = & 2 & - & x_3 & + & x_5 \\ \hline z & = & 3 & + & x_3 & - & 2x_5 \end{array}$$

# An Introductory Example cont'd

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*Can we still improve z??*

# An Introductory Example cont'd

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*Can we still improve  $z$ ??*



**Yes! :')**

# An Introductory Example cont'd

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**Repeat Step 5!:** Find the entering variable that will improve  $z$

The objective becomes  $z = 3 + x_3 - 2x_5$

# An Introductory Example cont'd

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Only improving  $x_3$  will improve  $Z$ ; thus, the entering variable is  $x_3$

# An Introductory Example cont'd

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The objective becomes  $z = 3 + x_3 - 2x_5$

Only improving  $x_3$  will improve  $Z$ ; thus, the entering variable is  $x_3$



How much can you increase  $x_3$ ?

$$x_1 = 1 + x_3 - x_5$$

$$x_2 = 2 - x_5$$

$$x_4 = 2 - x_3 + x_5$$

$$z = 3 + x_3 - 2x_5$$

Does not restrict!

$$x_3 = 2$$

# An Introductory Example cont'd

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**Step 5 cont'd:** Find the entering variable that will improve  $z$

The entering variable  $x_3 = 2$  while,  $x_1 = 3$ ,  $x_2 = 2$ ,  $x_4 = 0$ ,  $x_5 = 0$

***Basis becomes  $B = \{1, 2, 3\}$ , and  $z = 5$***

# An Introductory Example cont'd

**Step 6:** Update the simplex tableau

$$\begin{array}{rclcl} x_1 & = & 3 & - & x_4 \\ x_2 & = & 2 & & - x_5 \\ x_3 & = & 2 & - & x_4 + x_5 \\ \hline z & = & 5 & - & x_4 - x_5 \end{array}$$

*Can we still improve  $z$ ??*



***No, it's  
optimal!!***